

Deepak Sathyan

Curriculum Vitae

[arXiv](#) | [InspireHEP](#) | [Google Scholar](#) | [ORCID](#) | [Website](#)

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📍 Mitchell Institute for Fundamental Physics and Astronomy, Texas A&M University, College Station, TX

Experience

Postdoctoral Research Associate

Mitchell Institute for Fundamental Physics and Astronomy
Texas A&M University

College Station, TX
Sept 2024 – Aug 2027
3 years

Education

University of Maryland

Ph.D. in Physics

Thesis: Unifying Searches for New Physics with Precision Measurements of the W Boson Mass
Advisor: Dr. Kaustubh Agashe

College Park, MD
Aug 2018 – July 2024

Boston University

Bachelor of Arts in Physics, Summa Cum Laude with Honors

GPA: 3.9
Minor: Mathematics
Thesis: Muon (g-2): Searching for the Muon Electric Dipole Moment
Advisor: Dr. Robert Carey

Boston, MA
Aug 2014 – May 2018

Research Interests

Beyond the Standard Model searches with Collider Physics: LHC and muon collider phenomenology, Randall-Sundrum and baryon-number-violation models

Searches for sub-GeV Dark Matter: dark matter-neutrino interactions, dark photon and dark sector production, cosmic ray-dark matter scattering

Neutrino Physics: supernova neutrino attenuation and detection, open-system observables, neutrinophilic mediators, neutrino NSI

Seminars

9. HEP Seminar at Northwestern University

Cosmic Ray–Dark Matter scattering explanation of the Galactic Center Excess

Sept 2026

8. Fermilab Cosmic Physics Center Seminar

Cosmic Ray–Dark Matter scattering explanation of the Galactic Center Excess

Sept 2026

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| 7. CMS Exotica general meeting
Unification of Searches and Measurements: Probing BSM with the W boson mass measurement | May 2024 |
| 6. Seminar at Texas A&M
A New Purpose for the W-mass Measurement: Searching for New Physics via l + MET | Dec 2023 |
| 5. Seminar at UT Austin
A New Purpose for the W-mass Measurement: Searching for New Physics via l + MET | Dec 2023 |
| 4. CMS SUSY general meeting
A New Purpose for the W-mass Measurement: Searching for New Physics via l + MET | Nov 2023 |
| 3. HEP Seminar at University of Pittsburgh
A New Purpose for the W-mass Measurement: Searching for New Physics via l + MET | Nov 2023 |
| 2. HEP Seminar at Northwestern University
A New Purpose for the W-mass Measurement: Searching for New Physics via l + MET | Oct 2023 |
| 1. Theory Seminar at Washington University in St. Louis
A New Purpose for the W-mass Measurement: Searching for New Physics via l + MET | Oct 2023 |

Conference Talks

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| 14. Light Dark World 2026 at Carleton University [Invited]
Can the LHC be sensitive to the Light Dark World? | July 2026 |
| 13. CETUP* Workshop [Invited]
Producing the GeV Galactic Center Excess via Cosmic Ray-Dark Matter Scattering | June 2026 |
| 12. Particle Physics on the Plains at the University of Kansas [Contrib.]
A Baryon and Lepton Number Violation Model Testable at the LHC | Nov 2025 |
| 11. CETUP* Workshop [Invited]
Can the LHC be sensitive to light dark mediators? | June 2025 |
| 10. Phenomenology Conference at the University of Pittsburgh [Contrib.]
Can the LHC be sensitive to light dark mediators? | May 2025 |
| 9. Particle Physics on the Plains at the University of Kansas [Contrib.]
A comprehensive analysis of supernova neutrino-dark matter interactions | Nov 2024 |
| 8. Texas TACOS @ UT Austin [Invited]
A comprehensive analysis of supernova neutrino-dark matter interactions | Oct 2024 |

7. NuFact 2024 Conference at Argonne National Laboratory [Invited] A comprehensive analysis of supernova neutrino-dark matter interactions	Sept 2024
6. Mitchell Conference at Texas A&M University [Invited] A comprehensive analysis of supernova neutrino-dark matter interactions	May 2024
5. Particle Physics on the Plains at the University of Kansas [Contrib.] A New Purpose for the W-mass Measurement: Searching for New Physics via l + MET	Oct 2023
4. Phenomenology Conference at the University of Pittsburgh [Contrib.] Probing Dark Matter-Neutrino Interactions via Supernova Neutrinos	May 2023
3. Phenomenology Conference at the University of Pittsburgh [Contrib.] Model-Independent Measurement of Top Quark Mass Using B-Hadron Decay Lengths (Part I)	May 2022
2. April APS Meeting [Contrib.] Signals of KK graviton from extended warped extra dimensions at the LHC (II)	Apr 2021
1. Phenomenology Conference at the University of Pittsburgh [Contrib.] Signals of KK graviton from extended warped extra dimensions at the LHC (II)	May 2020

Teaching

Principles of Modern Physics Teaching Assistant, University of Maryland	Spring 2024
Electricity and Magnetism I Teaching Assistant, University of Maryland	Spring 2024
Mathematical Methods for Physics II Teaching Assistant, University of Maryland	Fall 2021, Fall 2022
Fundamentals of Physics I Teaching Assistant, University of Maryland	Summer 2019, Spring 2022
Introduction to Quantum Mechanics I (Graduate) Teaching Assistant, University of Maryland	Fall 2020
Fundamentals of Physics for Biologists II Teaching Assistant, University of Maryland	Spring 2019, Spring 2020
Fundamentals of Physics for Biologists Teaching Assistant, University of Maryland	Fall 2018, Fall 2019
Guest Lecture for Undergraduate E&M at University of Maryland Lectured on vector potentials in E&M	Mar 2024

Guest Lecture for Graduate E&M at Texas A&M University	Oct 2025
Lectured on relativistic Larmor formula and charged particle motion in magnetic fields	

Service

The Mitchell Conference on Collider, Dark Matter, and Neutrino Physics 2026
Conference Organizer — Texas A&M University, May 27-30, 2026
The Mitchell Conference on Collider, Dark Matter, and Neutrino Physics 2025
Conference Organizer — Texas A&M University, May 13-16, 2025
Mitchell Institute High Energy Phenomenology, Experiment, and Cosmology Seminar Series
Weekly Seminar Organizer - Texas A&M University, Fall 2024 - Present

Awards

Breakthrough Prize in Fundamental Physics	Apr 2026
Awarded to the Muon g-2 Collaboration; recognized as a collaboration member	
Ralph Myers & Friends of Physics Award	Dec 2019
Awarded by the University of Maryland Department of Physics	
Ralph Myers & Friends of Physics Award	May 2020
Awarded by the University of Maryland Department of Physics	
Alumni Award	May 2018
Awarded by the Boston University Department of Physics	

Schools Attended

GGI Lectures on the Theory of Fundamental Interactions	Jan 2023
TASI 2024: The Frontiers of Particle Theory	June 2024

Outreach

BU Student Mentorship Program PRISM	2016 - 2018
Mentored incoming freshman undergraduates in physics, providing guidance on courses, research opportunities, and resources	
BU Outreach for Local High Schools	Jan 2018
Demonstrated measurement of the charge-to-mass ratio of the electron for local high schoolers in physics	
TAMU Physics and Engineering Festival	2025 - present
Presented demonstrations of fluid dynamics to the general public, including high school students and families	

University of South Dakota talk: "My path through particle physics research so far"

Dec 2024

Presented my path through particle physics research to students, showing a variety of interesting ideas to pursue

Skills

Programming:

Mathematica, Python, C++

Physics Software:

MadGraph5_aMC@NLO, Pythia6, Pythia8, Delphes, ROOT, GEANT4, FeynRules

References

Dr. Kaustubh Agashe

University of Maryland, College Park — kagashe@umd.edu

Dr. Bhaskar Dutta

Texas A&M University — dutta@tamu.edu

Dr. Bhupal Dev

Washington University in St. Louis — bdev@wustl.edu

Dr. Jason Kumar

University of Hawai'i — jkumar@hawaii.edu